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Ketones in Blood

What is a ketones in blood test?

A ketones in blood test measures the level of ketones in your blood. Ketones are acids that your body makes when it breaks down fat for energy.

Normally, the cells in your body use **blood glucose** [<https://medlineplus.gov/bloodglucose.html>] , also called blood sugar, for energy. You get glucose from foods you eat, such as **carbohydrates** [<https://medlineplus.gov/carbohydrates.html>] . Your body breaks down most of that food into glucose and releases it into your bloodstream. If your cells can't get enough glucose, your body will break down fat for energy instead. This process produces an acid called ketones. If your body breaks down fat too fast, ketones can build up in your blood and urine.

High ketone levels make your blood too acidic. This is a serious condition called ketoacidosis. The most common type of ketoacidosis is a **complication of diabetes** [<https://medlineplus.gov/diabetescomplications.html>] called diabetic ketoacidosis (DKA). DKA is a medical emergency that often develops quickly and can lead to a coma and be life-threatening.

If you have diabetes, a ketones in blood test can help find high ketone levels early so you can get treatment right away. If your health care provider recommends frequent testing to check for increasing ketone levels, **at-home** [<https://medlineplus.gov/lab-tests/at-home-medical-tests/>] blood testing meters are available that check both glucose levels and ketones. At-home testing for ketones in urine [<https://medlineplus.gov/lab-tests/ketones-in-urine/>] testing may also be an option. Talk with your provider about which is right for you.

Other names: Ketone bodies (blood), serum ketones, beta-hydroxybutyric acid, acetoacetate, nitroprusside test

What is it used for?

With diabetes:

A ketones in blood test is mostly used to check for diabetic ketoacidosis (DKA) in people with diabetes. DKA can affect anyone with diabetes, but it is most common with type 1 diabetes [<https://medlineplus.gov/diabetestype1.html>] . With type 2 diabetes [<https://medlineplus.gov/diabetestype2.html>] , DKA is less common.

If you have diabetes, a blood ketones test may be used when you are sick or pregnant. You may also be tested if you develop diabetes during pregnancy [<https://medlineplus.gov/diabetesandpregnancy.html>] .

If you don't have diabetes, you may have your blood ketones checked if you are at risk for developing ketoacidosis and you have symptoms. You are at risk for ketoacidosis if you have a condition that causes your body to use fat for energy instead of glucose, such as:

- Eating disorders [<https://medlineplus.gov/eatingdisorders.html>] and starvation
- Vomiting [<https://medlineplus.gov/nauseaandvomiting.html>] or diarrhea [<https://medlineplus.gov/diarrhea.html>] that lasts a long time
- Alcohol use disorder [<https://medlineplus.gov/alcoholusedisorderaud.html>] (AUD)

Why do I need a ketones in blood test?

You may need a ketones in blood test if you have diabetes or other conditions that increase your risk for developing high ketone levels, or symptoms of ketoacidosis. You may also need the test if you have symptoms, and you eat a low-carbohydrate diet.

Symptoms of ketoacidosis may include:

- Extreme hunger or thirst
- Urinating (peeing) a lot more than usual
- Dehydration [<https://medlineplus.gov/dehydration.html>]
- Headache [<https://medlineplus.gov/headache.html>]
- Nausea and vomiting
- Pain in your abdomen (belly) [<https://medlineplus.gov/abdominalpain.html>]
- Confusion
- Trouble breathing
- Fatigue [<https://medlineplus.gov/fatigue.html>]
- Dry skin and mouth

- Breath that has a fruity smell
- Muscle stiffness or aches

If you don't get treatment, symptoms can quickly become more severe. If you have severe symptoms of ketoacidosis, seek immediate medical help. Large amounts of ketones can be life-threatening.

What happens during a ketones in blood test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

To check blood ketones at home, you may use a blood glucose meter that also tests for ketones. You will use a device to prick your finger. You'll collect a drop of blood on a ketone test strip that is inserted into the meter. The meter will show your ketone levels. Follow the instructions carefully. You can also test for ketones in urine using an at-home kit. Ask your provider how you should test for ketones, when you should test, and what to do if your levels are high.

Will I need to do anything to prepare for the test?

You don't need any special preparations for a ketones in blood test.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

A normal or negative test result usually means that your body is not making too many ketones and you don't have ketoacidosis. But a normal test result doesn't always rule out diabetic ketoacidosis (DKA). That's because your body makes three types of ketones. If DKA is in the early stages, only one type of ketone will be high, and lab tests often don't measure this type of ketone.

A high or positive test result means you have ketones in your blood. It's normal to have some blood ketones, but if your levels are very high and you have symptoms, you may have ketoacidosis. If you have diabetes, you likely have DKA. If you don't have diabetes, you may have ketoacidosis from alcohol use disorder, malnutrition [<https://medlineplus.gov/malnutrition.html>] , not eating for very long periods of time, or other conditions.

Ketoacidosis from any cause must be treated. Treatment may include going to the hospital.

If you have questions about your results, talk with your provider.

Learn more about laboratory tests, reference ranges, and understanding results [<https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/>] .

Is there anything else I need to know about a ketones in blood test?

People who eat a low-carbohydrate ketogenic ("keto") diet to lose weight will have blood ketone levels that are higher but are not high enough to be ketoacidosis. This condition is called ketosis.

Ketosis doesn't make your blood too acidic because your body is able to use the ketones, so they don't build up. But it's possible to develop ketoacidosis on a keto diet, so it's important to talk with your provider to see if it's safe for you.

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